Freight Rate Prediction - Write-up

Data

train_test.csv holds 48,000 labeled loads from 2025-01-01 to 2025-10-31 across 64 pickup and 64 delivery cities and 3 equipment types (Dry Van, Flatbed, Reefer). The target is posted_rate (USD). validation.csv covers 2025-11-01 to 2025-12-31 (12,000 rows) and is the final out-of-sample scoring set.

Data quality

- Negative / zero weights - 292 rows have weight <= 0 (minimum -47,500). Treated as missing and imputed with the training median (~31,436 lb).
- Missing values - weight missing in 300 train / 165 val rows, market_index missing in 374 train / 249 val rows. Both median-imputed; medians fit on training only to avoid leakage.
- Rate-per-mile outliers - 99.9th percentile of posted_rate is $12.8k with a max of $25.5k. Instead of clipping, the model is trained on log1p(posted_rate) with absolute-error loss so a few extremes don't dominate.
- Unseen cities in validation - ~6% of val rows have pickup or delivery cities not present in training. The gradient-boosted model handles these via haversine distance and lat/lon features, which carry most of the signal when the city label itself is unknown.

Features

```
| Feature | Why |
|---|---|
| distance | Correlation 0.91 with rate - the dominant driver. |
| haversine (from lat/lon) | Robust fallback when the city label is unseen; also catches cases where
| equipment | Rate per mile differs by equipment (Reefer $2.38 > Flatbed $2.29 > Dry Van $2.11). |
| weight | Minor signal, kept after cleaning. |
| market_index, quote_signal | Market/quote context. |
| pickup, delivery | Native categoricals - captures lane effects beyond distance. |
| dayofweek, dayofyear, month, week | Seasonal and weekday effects. |
```

Model

sklearn.ensemble.HistGradientBoostingRegressor with absolute-error loss, native categorical handling, and the target on log scale. Chosen over LightGBM / XGBoost because:

- No external native dependency (no libomp on macOS) - easy to reproduce.
- Native categorical support, so no high-cardinality one-hot blow-up for the 64 cities.
- Absolute-error loss plus log1p target is robust to the few $20k+ outliers.

Key hyperparameters: learning_rate=0.05, max_leaf_nodes=63, min_samples_leaf=40, max_iter=1200, early stopping on 10% internal validation when fitting the holdout model.

Validation approach

Because validation.csv is the future (Nov-Dec vs. a Jan-Oct training window), a time-based holdout is the only honest split. The last 30 days of training (2025-10-02 to 2025-10-31) are the holdout; the first ~9 months are the fit set.

Holdout metrics (log target, metrics reported in dollars):

```
| Metric | Value |
|---|---|
```

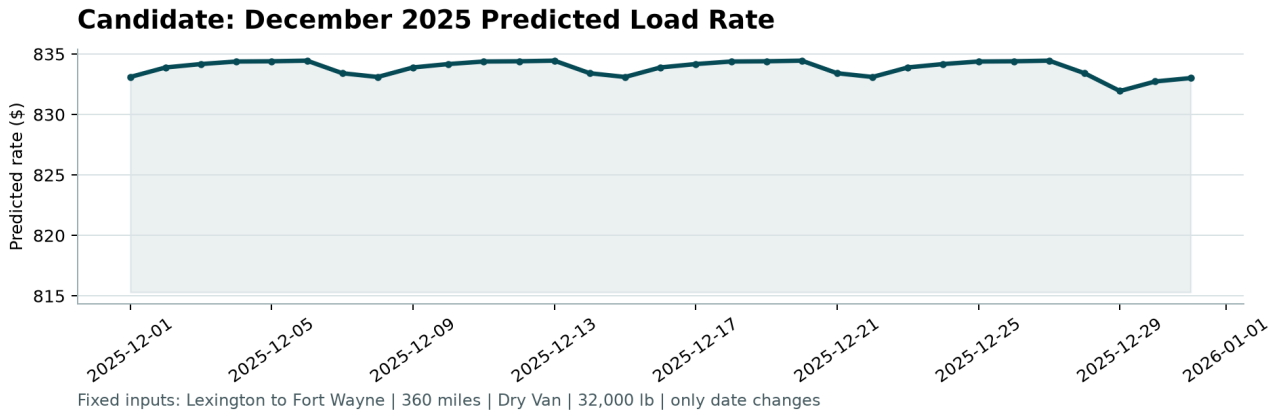
| MAE | $101.22 |
| MAPE | 4.87% |

Random k-fold would leak future information into the training folds and overstate performance, so I didn't use it.

Once the holdout number was accepted, the final submitted model was refit on all 48,000 rows with the same hyperparameters.

December chart

Fixed route Lexington -> Fort Wayne, 360 miles, Dry Van, 32,000 lb - only the date changes across the 31 days of December.



Predictions sit around $833 (~$2.32/mi), close to the training average for the Lexington -> Fort Wayne lane ($856, 32 samples). Day-to-day movement is small and comes from the dayofweek effect learned from training; raw ordinal date features (dayofyear, week, month) can't extrapolate past October in a tree model, so the shape is intentionally smooth rather than invented trend.

One subtle fix worth calling out: the dataset's pickup_lat/pickup_lon for each city are not the real geographic coordinates - they're the dataset's own fixed per-city values. For the fixed December route I pull them straight from the training file so the haversine feature matches what the model saw during training; using Google-maps coords put December in the wrong leaf and dropped the baseline by ~$40.

Reproducing

```
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python -m src.train
python -m src.predict
python score.py --predictions validation_predictions.csv --december-predictions data/december_c
```